AgNW FDT Memristor Framework

Simulator of the stochastic silver-nanowire (AgNW) memristor network of Lamas et al. (2026), plus a fluctuation-dissipation test (FDTest) for memory (non-Markovianity) in its dynamics. Part of a broader project connecting computational neuroscience, neuromorphic hardware, and non-equilibrium statistical physics.

Where to start reading the science: `docs/REPORT_FDTest_progress_brief.md` (collaborator-facing, no repo-internal references) or `docs/REPORT_FDTest_progress.md` (full, internal). PDFs of both are built with `docs/md_to_pdf.sh`. `AgNW_FDT_Memristor_Framework.md` is the original theory document that motivated this work (largely superseded by the FDTest approach — see the report's §5.1). Current switching equations: `docs/switching_model.md`. Theory audit: `docs/theory_audit/THEORY_AUDIT.md`.

> **Legend (color code, fixed hexes).** **green** = worked / pass · **red** = failed / fail · **blue** = future work / planned test · **violet** = parameter still to be determined (measure or ask authors). Color is emphasis only — every verdict also says **pass**, **fail**, **planned**, or **unconfirmed** in words.
>
> **30-second jargon. AgNW network** (random sticks = wires; crossings = memristor junctions) · **SET/RESET** (voltage-driven OFF $\rightarrow$ ON / current-driven ON $\rightarrow$ OFF switching) · $g_{\mathrm{eff}} = I_{\mathrm{total}}/V_{\mathrm{in}}$ (whole-network conductance, our main observable) · **FDTest** (perturbation test for memory: kick the voltage, compare relaxation vs. spontaneous fluctuations) · Δ^2 (its violation score: 0 = memoryless/**Markov**, clearly above noise floor = memory/**non-Markov**) · M (averaged perturbation cycles, ≈ 1000) · S/N (response $\div$ noise; calls need $S/N \geq 3$).

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Methodology

Fixed simulator. Random stick network (wires of length $70\,\mu$m dropped with random position/orientation in a $1\,\mathrm{mm}^2$ square; crossings form **junctions**; wires in a 5% edge band are the electrodes; $N = 2000$ wires $\rightarrow$ ≈ 6000 junctions), **dual-node Kirchhoff circuit** per Lamas et al. §2.2 (each junction = two nodes joined by $G_j \in \{G_{\mathrm{OFF}}, G_{\mathrm{ON}}\} = \{10^{-9}, 10^{-3}\}$ S, wire segments resistive with $\rho_{\mathrm{Ag}} = 1.59 \times 10^{-8}\,\Omega$m and diameter D), **stochastic junction dynamics** stepped at $\Delta t = 1$ ms. Import `agnw_sim` and never copy it: `from agnw_sim.dynamics import SimParams`, `from agnw_sim.simulate import run_simulation`.

Junction dynamics (working model). With $V_{\mathrm{mem}} = |V_a - V_b|$ (voltage drop across the junction edge) and $I_{\mathrm{mem}} = |G_j V_{\mathrm{mem}}|$ (current through it):

$$P_{\mathrm{SET}} = \Delta t\, P_0\, [1 - e^{-\alpha(V_{\mathrm{mem}} - V_{\mathrm{th}})}]_+, \qquad P_{\mathrm{RESET}} = \min(1,\, \Delta t\, P_{\mathrm{decay}}\, (I_{\mathrm{mem}}/I_{\mathrm{unit}})^2),$$

plus **spontaneous ON $\rightarrow$ OFF relaxation** (each ON junction turns OFF with probability p_{relax} /step, independent of voltage/current/history; recalled from a Lamas talk, not in the paper).

Withdrawn/superseded — see corrected rule below. Before 2026-09-19 the sim treated Table 1's 0.4 / 0.01 as *rates* and multiplied by Δt again, making SET $1000\times$ too slow (the $P_{\mathrm{decay}} \approx 3 \times 10^8$ finding and avalanche table from that period are invalid). **Correction:** Table 1 lists *per-step products* $\Delta t\, P_0 = 0.4$, $\Delta t\, P_{\mathrm{decay}} = 0.01$ — always build with `SimParams.from_step_probs(0.4, 0.01)`. Anything under `validation_output/old_model/` or `docs/on_hold/validate_p_decay/` uses the mis-scaled model — do not run or build on it.

Fig. 4 calibration. The paper's Fig. 4 I-V hysteresis (**sine drive**, 3.6 V amplitude, 50000 steps/cycle, at **network sizes** $N = 2000/3000$) is the target; metrics are $I_{\max}$ (peak current: paper 3.24 mA at $N = 2000$), **onset** (first voltage with $I > 0.05$ mA: paper ≈ 1.9 V), **loop areas** $A_{\pm}$ (positive/negative half-cycle $\oint I\,dV$: paper $A_+ = 2.96$ mA V, $A_-/A_+ = 0.44$ at $N = 2000$). Four quantities could not be used as printed and were inferred by matching Fig. 4 (see table below).

FDTest engine. The theory document's X (response/correlation ratio) is unmeasurable here (no $k_B T$, no voltage-conductance coupling; several of its equations are wrong — see theory audit). Instead the **FDTest violation** Δ^2 (Geli, Monti et al.; same measure as the brain-dynamics side of the project): kick the voltage by **perturbation** ε ($\varepsilon \times 1 \approx 13$ mV, $\times 3 = 40$ mV, $\times 6 = 80$ mV), compare relaxation vs. spontaneous autocorrelation over 1400 lags; $\Delta^2 > 0$ =

memory. **OFF/ON/BOTH** = switch-off mirror / switch-on mirror / average estimators. Protocol: fixed-voltage pilots to pick the **operating point** V_0 (baseline voltage, 2.5 V), chained-cycle campaign ($T_{\mathrm{relax}} = 30000$, $T_{\mathrm{samp}} = 15000$; M = total cycles $= 12 \times n_{\mathrm{seeds}}$), analysis (`fdtest_geff` light / `fdtest_analyze` full / `fdtest_observables`), Monte Carlo decision-rule calibration, acceptance gate, **null control** (no-perturbation run: $\Delta^2 \approx 1$ by construction, gate must stay silent).

Validation. Exact two-state/renewal benchmarks validate the estimator; a direct memory diagnostic on $g_{\mathrm{eff}}(t)$ and an exactly enumerable 11-junction toy circuit cross-check the violation; Δ^2 vs. M out to $M = 2016$ confirms saturation.

Switching variants: what was tried, what works

Self-contained: each row states the variant, what it means in plain words, and the tagged verdict — no need to open the report first.

variant	in plain words	result (with conditions)	verdict
SET, $V_{\mathrm{th}} = 0.01$ V (as printed)	junctions switch ON almost as soon as any voltage appears	onset near 0.35 V vs. paper 1.9 V ($N = 2000$, RESET off); only the V_{th} sweep moves the onset	**fail** — printed value wrong
SET, $V_{\mathrm{th}} = 0.1$ **V (inferred, unconfirmed)**	junctions need a sizable local drop before switching	onset ≈ 1.55 V; same value works at $N = 3000$ (gap 1.55 vs. 1.9 V still open)	**unconfirmed fit** — brackets paper at both sizes
Ideal wires ($D = 0$)	wires conduct perfectly, only junctions resist	≈ 1.2–$1.3\times$ too much current ($N = 2000$)	**fail** — resistance needed
Wire diameter $D \approx 80$–90 nm (inferred, unconfirmed)	thin wires add series resistance, lowering current without moving onset	peak current crosses paper's 3.24 mA at $D \approx 80$–90 nm ($N = 2000$)	**unconfirmed fit** — diameter never stated
RESET literal ($\Delta t\, P_{\mathrm{decay}} = 0.01$)	current-driven ON → OFF with the paper's	$A_-/A_+ = 0.15$ vs. 0.44; nearly	**fail** — cannot make the

variant	in plain words	result (with conditions)	verdict
	coefficient	inactive at Fig. 4 currents ($N = 2000$)	negative loop
RESET scaled $3\times$ –$300\times$	stronger current-driven turn-off	never the right loop with the right currents	**fail** — systematically wrong
Spontaneous relaxation $p_{\mathrm{relax}} = 10^{-4}$/step (working model)	ON junctions spontaneously decay (mean life 10^4 steps ≈ 10 s), the missing negative-loop mechanism	$A_-/A_+ = 0.53$ vs. 0.44 ($N = 2000$, 4 seeds), 0.56 vs. 0.49 ($N = 3000$); stationary in $\sim 10^3$ steps at fixed V	**pass** (mechanism itself **unconfirmed**: talk, not paper)
$p_{\mathrm{relax}} = 3\times10^{-5}$ / 2×10^{-4} / 10^{-3}	weaker / stronger / collapsing decay	$A_-/A_+ = 0.34$ / 0.68 / 0.99 ($N = 2000$); 10^{-3} collapses the current	**fail** as working values — brackets only, 10^{-4} kept
Fixed lifetime $L = 20000$	every ON junction dies exactly L steps after SET	fits loops ($A_-/A_+ = 0.54$) but population collapses synchronize at multiples of L at fixed V (cyclostationary)	**fail** as working model — unphysical for FDTest
Gamma lifetime $k \in \{1, 2, 4, 8, 16\}$, mean L (planned)	lifetime family joining exponential ($k = 1$) to fixed ($k \to \infty$); a thinning filament should have increasing hazard, i.e. intermediate k	not yet run; expects smooth metric variation, best k = inferred shape	**planned** next — cheapest candidate mechanism
Multi-level filaments (planned)	filaments form in stages, not clean binary switch	not yet implemented; would make g_{eff} memory real, not a projection artefact	**planned** — most interesting, most expensive

variant	in plain words	result (with conditions)	verdict
Bipolar RESET (deferred)	reset only under reverse bias, not I^2 regardless of sign	needs literature/data check first; touches RESET everywhere	**planned** — deferred
Thermal cross-talk (deferred)	heat diffusion couples neighbouring junctions = real spatial memory	breaks the independent-junction speed assumption	**planned** — most expensive, deferred
Random telegraph noise (lowest priority)	formed-filament flicker, cheap to add	no clear link to the memory question	**planned** — lowest priority
Theory-doc X (Eqs. 7,8,10,12–14,16)	original response/correlation ratio	wrong sign, wrong g-dependence, no $1/N$ (audit findings F3/F6/F10)	**fail** — superseded by FDTest Δ^2

Working preset: `agnw_sim.presets.lamas2026_fig4` + $p_{\text{relax}} = 10^{-4}$ + $D = 90$ nm. Open with authors: I_{unit} (1 mA assumed), real RESET activity, relaxation form, onset offset, Fig. 5A plateau (see `docs/lamas_questions_es.md`).

Headline results

Fig. 4 replication ($N = 2000$ 4 seeds, $N = 3000$ 1 seed; sine 3.6 V). **Pass**: $V_{\text{th}} = 0.1$ V + $D \approx 90$ nm + $p_{\text{relax}} = 10^{-4}$ reproduces shape, size, and asymmetry without retuning across sizes. Remaining offsets: onset 1.55 vs. 1.9 V, I_{max} +5–10%; Fig. 4 and Fig. 5A share no single parameter set yet (**unconfirmed** with authors).

FDTest on g_{eff} ($N = 2000$, $V_0 = 2.5$ V, seed 1). **Pass (non-Markov)**: $\Delta^2 = 0.10$ $[0.02, 0.13]$ vs. floor 0.018 at $\varepsilon \times 3$ ($M = 924$, $S/N \approx 9$–15); 0.14–0.30 with switch-on included; $\varepsilon \times 6$ ON 0.42 $[0.31, 0.50]$. Null control $\Delta^2 \approx 0.97$–0.99, correctly no call (**pass**). Δ^2 flat at 0.10–0.12 from $M \approx 600$ through $M = 2016$ — saturated, not data-starved (**pass**). Junction-class Δ^2 (0.03–0.1) is sampling-noise artefact (same with no perturbation). ON fraction and cross-fitted linear score carry no response. Two qualifications (see Limitations): one gate check **fails** (odd/even 0.174 vs. 0.106 — likely cycle 0 in the even half); switch-on $\approx 2\times$ switch-off at every ε, unexplained.

What the violation does/does not show. $\Delta^2 > 0$ says g_{eff} **is non-Markov** (its relaxation differs from its autocorrelation by 30–60% RMS) — not that the dynamics have memory (the full

junction state is exactly Markov). Two cross-checks: toy-circuit coarse-graining reaches only 0.013 vs. measured 0.10 (not enough); a direct memory diagnostic on real $g_{\text{eff}}(t)$ finds real memory at every lag, same order as Δ^2, with clean nulls on memoryless synthetics. Remaining candidates: size effect or a physical mechanism (Gamma relaxation cheapest to test).

Figures: what, how, so-what

All in `docs/report_figures/` (committed PNGs), generated by `make_figures.py` / `make_dynamics_figures.py`.

- `network_N2000.png` — **What:** the $N = 2000$ network (5953 junctions); left whole network, right 0.1 mm zoom; red = V_{in} electrode wires, blue = ground wires, black dots = junctions. **How:** check orientations look random, electrodes only at edges, junctions wherever wires cross. **So-what:** this is the exact geometry all FDTest runs use (network seed 1).
- `dyn_conductance_traces.png` — **What:** conductance of 20 of 1000 recorded junctions over 60 s (60000 steps) at constant $V_{\text{in}} = 2.5$ V, all-OFF start, $G_{\text{OFF}} = 10^{-9}$ S / $G_{\text{ON}} = 10^{-3}$ S. **How:** flat lines = idle, jumps = SET/RESET/relaxation events. **So-what:** junctions flip rarely and stay put for seconds — the network has slow elements (**pass**: the timescale the FDTest needs exists).
- `dyn_current_trace.png` — **What:** left total current after switching on $V_{\text{in}} = 2.5$ V (60 s); right g_{eff} over the last 30 s. **How:** look for the rise ending and the trace going flat. **So-what:** current rises ~ 10 s then fluctuates around a flat level — that flat part is the stationary state the FDTest requires (**pass**).
- `dyn_autocorrelation.png` — **What:** left ensemble ON/OFF-state autocorrelation (stationary part); right normalized g_{eff} autocorrelation. **How:** fast initial drop vs. slow tail. **So-what:** two timescales (fast tens of steps + slow tail), consistent with the rare-switching junctions above.
- `dyn_network_state.png` — **What:** junction state after 60 s at 2.5 V; red = ON, grey = OFF; 311/1000 recorded junctions (31%) ON. **How:** count the red fraction. **So-what:** the FDTest operating point is well into the conducting regime (saturated switching, not onset) — a generalizability caveat (see Limitations).
- `onset_sweep.png` — **What:** conduction onset (first V with $I > 0.05$ mA) for SET-parameter variations at $N = 2000$, RESET off; baseline ($V_{\text{th}} = 0.01$ V) vs. sweeps; red dashed = paper (≈ 1.9 V). **How:** only the V_{th} curve reaches the red line; $V_{\text{th}} = 0.5$ V never conducts. **So-what:** the printed 0.01 V must be wrong (**fail**); $V_{\text{th}} = 0.1$ V brackets the paper (**pass**, **unconfirmed** with authors).
- `wire_diameter_sweep.png` — **What:** peak rising-branch current vs. wire diameter D ($N = 2000$; $D = 0$ = ideal wires); red dashed = paper 3.24 mA. **How:** the curve crosses the

red line at $D \approx 80$–90 nm; onset does not move. **So-what:** the two calibrations are independent (**pass**; **unconfirmed** — diameter never stated).

- `relaxation_scan.png` — **What:** A_-/A_+, A_+, $I_{\max}$ at $N = 2000$ (purple) vs. p_{relax}; red dashed = paper; green square/orange dotted = $N = 3000$ at 10^{-4} and paper's $N = 3000$ ratio; grey line = working 10^{-4}. **How:** the asymmetry ratio climbs with relaxation strength while $I_{\max}$ stays put; 10^{-4} is where purple meets red at both sizes. **So-what:** 10^{-4} fits at $N = 2000$ and $N = 3000$ (**pass**; mechanism **unconfirmed**).
- `gallery_seeds_N2000.png` / `gallery_N3000.png` — **What:** simulated loops (purple) vs. digitized paper (red): 4 seeds at $N = 2000$ (top fixed $L = 20000$, bottom exponential 10^{-4}), 1 seed at $N = 3000$; paper $A_-/A_+ = 0.44/0.49$, $I_{\max} = 3.24/10.66$ mA. **How:** shape and size match at every seed. **So-what:** calibration is not a one-seed accident (**pass**); loops alone cannot tell exponential from fixed lifetime.
- `fixed_v_overview.png` — **What:** long constant-V_{in} runs (60000 steps, $V_{\text{th}} = 0.1$ V); left G_{eff}, right ON fraction; rows: literal RESET / exponential / fixed lifetime / last two at $N = 2500$. **How:** top row never flattens, second flattens in $\sim 10^3$ steps, third crashes periodically. **So-what:** literal RESET gives no stationary state (**fail** $\rightarrow$ no FDTest); exponential is stationary (**pass** $\rightarrow$ FDTest uses it); fixed lifetime is unphysical (**fail**).
- `fdtest_protocol.png` — **What:** chained-cycle schedule with a real $g_{\text{eff}}(t)$ trace (seed 1, mid-run stationary cycles) and the M-cycle ensemble average. **How:** the perturbation is invisible in one trace, clear in the average. **So-what:** this is why the FDTest must average over many cycles.
- `geff_switch_eps3_M924.png` — **What:** g_{eff} at $\varepsilon = 40$ mV, $M = 924$; left switch-off, right switch-on; blue = response, orange = correlation, grey = noise, red/black dashed = 5% and noise-based cutoffs. **How:** blue visibly separates from orange both ways. **So-what:** the memory signal, $\Delta^2 = 0.10$ / 0.18 (**pass**).
- `geff_switch_null.png` — **What:** same with no perturbation ($M = 220$). **How:** both curves are grey-level noise. **So-what:** control behaves correctly, $\Delta^2 \approx 1$, gate silent (**pass**).
- `fdtest_geff_eps3_M1008.png` — **What:** full-analysis response/correlation curves, $\varepsilon = 40$ mV, $M = 1008$, 1400 lags; grey line = cutoff τ_c. **How:** response above correlation out to the grey line, then both in noise. **So-what:** same gap with the full campaign (**pass**).
- `geff_delta2_vs_m.png` — **What:** $\Delta^2(g_{\text{eff}})$ vs. M (log-log) to $M = 2016$; dashed = $1/M$-scaled floor. **How:** steep drop to $M \approx 600$ (0.80 $\rightarrow$ 0.11), then flat 0.10–0.12; floor stays well below. **So-what:** a still-falling curve would mean "need more data"; this one is flat — 0.10 is the answer, not an upper bound (**pass**).
- `fdtest_benchmarks.png`, `markov_tests_tiny.png`, `markov_tests_geff.png` — **What:** estimator vs. exact two-state/renewal curves; toy-circuit Δ^2 (full state exactly 0, ON-count 0.013); direct diagnostic on real $N = 2000$ data at every lag + memoryless-synthetic nulls. **How:** estimator hugs the exact curves; toy 0.013 $\ll$ measured 0.10; diagnostic finds

memory at every lag, nothing on synthetics. **So-what:** estimator validated and two artefact explanations ruled out (**pass**).

Limitations

- **Unconfirmed (needs authors):** everything rests on the working relaxation model and inferred V_{th}, D — none confirmed by the authors.
- **Planned (statistical power):** $M \approx 500$–2016 here vs. ≈ 5000 in the brain-dynamics analogue — a smaller real effect could go undetected.
- **Planned (generalizability):** one network, one operating point (2.5 V, near-saturated switching); untested at 1.6 V near onset or other seeds; regime comparison and network-level controls (rewired nulls, isolated-junction) are future work.
- **Fail (acceptance gate):** odd- vs. even-numbered cycles give inconsistent Δ^2 (0.174 vs. 0.106) — likely cycle 0 (empty-network start) sitting in the "even" half; not yet re-confirmed with cycle 0 excluded.
- **Fail-or-bias (switch-on/off asymmetry):** switch-on Δ^2 consistently $\approx 2\times$ switch-off at every ε tested, no identified mechanism — real physics or analysis bias.
- Noise floor covers only $\langle Z \rangle$ noise and runs $\approx 10\%$ low against a Markov Monte Carlo; the Δ^2 vs. M curve (flat at 0.10–0.12 from $M \approx 600$) bounds this.
- Slow junctions (relaxation up to 10^4 steps) are only partly covered by the 14000-step window.
- A direct response/correlation cross-check (voltage-step nudge, watch g_{eff} relax) is **planned** as an independent check.

Status: where we are / doing / next

- **Where we are:** simulator + FDTest campaign done at one operating point ($N = 2000$, $V_0 = 2.5$ V): g_{eff} is non-Markov ($\Delta^2 = 0.10$, saturated to $M = 2016$, null control clean), Fig. 4 reproduced at two sizes with the working model.
- **What we are doing:** **planned next:** Gamma-lifetime scan ($k \in \{1, 2, 4, 8, 16\}$, mean $L = 20000$) as the cheapest physical-memory candidate, or a mid-size Monte Carlo bridge for the size-effect explanation (Juan's call).
- **What to do next + data asks:** **planned:** exclude cycle 0 and re-run the gate; explain ON $>$ OFF; operating point near onset (1.6–1.8 V); fixed-lifetime network positive control; more $\varepsilon \times 6$ statistics; $N = 1500/3000$ size dependence; $S/N \geq 3$ gate in analyzers. **Ask authors:** I_{unit}, real RESET activity, relaxation form, V_{th}, D, Fig. 5A plateau (see `docs/lamas_questions_es.md`).

Install

Requires Python ≥3.10.

```
python3 -m venv .venv
.venv/bin/pip install -e ".[dev]"
```

Dependencies: `numpy`, `scipy`, `h5py`, `matplotlib` (runtime), `pytest` (dev). See `pyproject.toml`.

Run the test suite with the venv's Python, not a bare `pytest` — `scripts` needs to be importable as a package:

```
.venv/bin/python -m pytest -q
```

(291 tests as of this writing, ~60 s.)

Repository layout

```
agnw_sim/        the installable package: circuit, dynamics, simulator, FDTest estimator
scripts/         standalone CLI drivers (python -m scripts.<name>), not imported by the package
tests/           pytest suite for agnw_sim/ and scripts/
docs/            theory document, progress reports, design specs, figures, results
validation_output/   git-ignored: everything scripts/ writes (run data, figures, JSON) — fully
regenerable
memory/, RESUME_FROM_HERE.md, docs/superpowers/   Claude Code session-continuity files (see
below)
```

Package: `agnw_sim/`

`agnw_sim/__init__.py` re-exports nothing — always import submodules directly, e.g. `from agnw_sim.dynamics import SimParams`, `from agnw_sim.simulate import run_simulation`.

module	purpose	key names
`geometry.py`	random stick network generation, wire intersections	`Network`, `build_network(n_wires, area, length, rng, ...)`
`circuit.py`	sparse Kirchhoff solve on the plain (non-dual-node)	`solve_circuit`, `total_current`,

module	purpose	key names
	graph	`junction_currents`
`dualnode.py`	the dual-node circuit of Lamas et al. Sec. 2.2 (each junction $\rightarrow$ two nodes + resistive wire segments)	`DualNodeCircuit`
`cached_solver.py`	opt-in solver acceleration (rescale/rank-1 update instead of a full refactorization every step)	`CachedSolver` (6.4x faster at stationarity)
`dynamics.py`	stochastic SET/RESET/spontaneous-relaxation junction switching (described in `docs/switching_model.md`)	`SimParams`, `p_set`, `p_reset`, `update_dynamics`, `update_dynamics_lifetime`
`presets.py`	named parameter sets matched to the paper's Fig. 4	`lamas2026_fig4(dt=1e-3)`
`waveforms.py`	drive-voltage generators	`triangle`, `sine`, `step`, `ramp`, `white_noise_probe`
`simulate.py`	the time-stepping loop	`run_simulation(...)`, `SimulationResult`
`io.py`	HDF5 trajectory storage	`save_result`, `load_result`, `reconstruct_network`
`visualize.py`	sanity-check plots	`plot_network`, `plot_iv_hysteresis`, `plot_conductance_trace`
`analysis.py`	trajectory analysis helpers	`autocorrelation`, `hysteresis_loop_area`, `entropy_production`
`benchmarks.py`	exact FDTest curves for analytically solvable junctions, used to validate the estimator	`two_state_fdtest`, `renewal_fdtest`

module	purpose	key names
`fdtest.py`	**the FDTest estimator** (pure functions)	`select_cutoff`, `delta_squared`, `binary_fdtest`, lag-binning helpers
`fdtest_protocol.py`	chained-cycle schedule, stage extraction, pilot analysis, acceptance-gate checks	`Schedule`, `extract_stage_arrays`, `choose_v0` / `choose_epsilon`, `gate_*`
`fdtest_observables.py`	alternative observables (ON fraction, cross-fitted linear score)	`on_fraction`, `fit_lda_direction`, `crossfit_scores`
`exact_chain.py`	exact one-step transition data of a small circuit (every 2^J state solved)	`dynamic_junctions`, `state_data`, `bridge_network`
`markov_tests.py`	memory diagnostics (regression gain, kNN conditional mutual information) and exact finite-chain FDTest quantities	`regression_memory_gain`, `cmi_test`, `transition_matrix`, `exact_delta2`

Drivers: `scripts/`

Run every driver as `python -m scripts.<name>` from the repo root, with `.venv/bin/python`. Each script's own docstring (`python -m scripts.<name> --help`, or just read the top of the file) has the full argument list; this is a map of what each one is for.

script	what it does
`fdtest_pilot.py`	fixed-voltage pilot runs, to choose the FDTest operating point (V_0)
`fdtest_run.py`	generates one seed's worth of chained FDTest cycles — the actual simulation
`fdtest_analyze.py`	full analysis: g_{eff} + per-junction classes, bootstrap intervals, noise floors, acceptance gate, figures.

script	what it does
	Memory scales with $M \times K$ (junctions); needs ~9 GB at $M = 480$, $K = 1000$
`fdtest_geff.py`	lighter g_{eff}-only analysis (switch-on mirror + noise-based cutoff), no per-junction memory cost
`fdtest_observables.py`	ON-fraction and cross-fitted linear-score analysis
`fdtest_interval_calibration.py`	Monte Carlo calibration of the decision rule against synthetic junctions with a known answer
`fdtest_tau_slow.py`	slowest per-junction relaxation time from a pilot run
`fdtest_common.py`	*not a driver* — shared setup helper reused by the other `fdtest_*` scripts
`geff_delta2_vs_m.py`	$\Delta^2(g_{eff})$ vs. number-of-cycles M convergence curve, reusing already-recorded runs (no re-simulation)
`fixed_v_stationarity.py`	long fixed-voltage runs, to check a model variant reaches a stationary state
`replicate_lamas_fig4.py`	reproduces the paper's Fig. 4 I-V hysteresis loops under a sine drive
`explore_fig4_onset.py`	SET-parameter sweep, used to diagnose the conduction-onset mismatch against the paper
`explore_wire_resistance.py`	wire-diameter sweep (found $D \approx 90$-100 nm)
`tiny_circuit_exact.py`	exact FDTest of a tiny, fully enumerable circuit (the "Markov test 2" of the report)
`markov_geff_test.py`	direct memory diagnostic on real $g_{eff}(t)$ data (the "Markov test 4" of the report)

`docs/`

path	contents
`REPORT_FDTest_progress.md` / `.pdf`	full progress report — the primary internal reference for results and methods

path	contents
`REPORT_FDTest_progress_brief.md` / `.pdf`	collaborator-facing version: no repo-internal file/script references, plain prose
`report_figures/`	committed PNGs used by both reports, plus the `make_figures.py` / `make_dynamics_figures.py` generators
`results/`	analysis outputs meant to be kept (unlike `validation_output/`): `interval_calibration.md`, `markov_tests.md`
`theory_audit/`	audit of the original theory document + Fig. 4 replication analysis; has its own README
`on_hold/optimization_framework/`	an optimization-framework implementation, parked because its core formula is unvalidated
`on_hold/validate_p_decay/`	a validation script moved out of `scripts/`: it uses the pre-fix mis-scaled model (double-applied Δt); do not run it
`superpowers/{specs,plans}/`	design specs and task plans from past work sessions — a paper trail of *why*, not needed to run anything
`switching_model.md`	the current junction switching model (SET, RESET, spontaneous relaxation): equations, per-step update, working-model parameters and their provenance
`lamas_questions_es.md`	draft list of open questions for the paper's authors
`lag_binning_design.md`, `alt_observables_design.md`	design docs for two implemented analysis extensions
`response_correlation_options.md`	a deferred design doc (not implemented)
`md_to_pdf.sh`	builds a report PDF from its Markdown source — see below

`tests/`

Standard pytest layout, one file per package module plus a few smoke tests for the `scripts/` drivers. Run with `.venv/bin/python -m pytest -q`.

Running a simulation

```
import numpy as np
from agnw_sim.presets import lamas2026_fig4
from agnw_sim.waveforms import sine
from agnw_sim.simulate import run_simulation

params, extra = lamas2026_fig4(dt=1e-3)          # SimParams, {"wire_diameter": 90e-9}
# waveform_fn takes the simulation time t (seconds) and returns V_in(t); waveforms.py's
        functions
# (sine, triangle, step, ramp) take t as their first argument, so wrap the fixed ones in a
        lambda:
waveform_fn = lambda t: sine(t, amplitude=3.6, frequency=1 / (50_000 * params.dt))

result = run_simulation(
    n_wires=2000, area=1e-6, length=70e-6,        # 1 mm^2 square, 70 um wires
    waveform_fn=waveform_fn, n_steps=50_000, dt=params.dt, params=params,
    rng=np.random.default_rng(1), wire_diameter=extra["wire_diameter"],
)
# result.I_total, result.G_sampled, ... — see agnw_sim.simulate.SimulationResult
```

For a constant-drive run instead, use `lambda t: step(t, amplitude=V0, t_on=0)` (from `agnw_sim.waveforms`). Add `solver="cached"` to `run_simulation(...)` for the 6.4x-faster solver (dual-node graph only, i.e. `wire_diameter` set) once the network has largely stopped switching.

Running the FDTest

The FDTest needs a *campaign* of chained-cycle runs at one operating point, then an analysis pass over them. This is the exact, previously-validated sequence (see `RESUME_FROM_HERE.md` for the full history):

```
# 1. Pick the operating point (V0) with fixed-voltage pilots at a few candidate voltages...
for v in 1.4 1.6 1.8 2.0 2.5; do
    .venv/bin/python -m scripts.fdtest_pilot run 2000 "$v" --n-steps 60000
done
# ...then compare their mean g_eff, drift and ON-probability spread:
.venv/bin/python -m scripts.fdtest_pilot summarize 2000

# 2. Generate the campaign: one call per seed (repeat for as many seeds as you want M = 12 *
      n_seeds cycles)
.venv/bin/python -m scripts.fdtest_run 2000 --v0 2.5 --eps 0.04 --t-relax 30000 --t-samp 15000 \
    --cycles 12 --seed 1 --n-lags 600 --out validation_output/my_run
# ... --seed 2, --seed 3, ... (independent processes; run many in parallel)

# 3. Analyze
.venv/bin/python -m scripts.fdtest_geff 2000 --out validation_output/my_run --n-boot 200 --n-
      lags 1400
.venv/bin/python -m scripts.fdtest_analyze 2000 --out validation_output/my_run --n-lags 1400
.venv/bin/python -m scripts.fdtest_observables 2000 --out validation_output/my_run --n-lags 1400
```

`fdtest_analyze.py` is the full analysis (network + per-junction classes, acceptance gate); `fdtest_geff.py` is a lighter g_{eff}-only pass with no per-junction memory cost — use it first, or when `fdtest_analyze.py`'s RAM footprint at your M doesn't fit. All three read the same run files under `--out` and never re-simulate.

To calibrate the decision rule itself (percentile vs. reverse-percentile intervals, false-call rates):

```
.venv/bin/python -m scripts.fdtest_interval_calibration --reps 60 --Ms 120,240,480 --procs 3
```

Other analyses

```
# Fig. 4 I-V hysteresis replication (lamas_epsE = exponential relaxation, probability E per
        step; the working value is 1e-4)
.venv/bin/python -m scripts.replicate_lamas_fig4 2000 lamas_eps1e-4 1

# Fixed-voltage stationarity check (same VARIANT tokens as replicate_lamas_fig4.py)
.venv/bin/python -m scripts.fixed_v_stationarity 2000 lamas_eps1e-4 2.5

# Exact tiny-circuit FDTest (Markov test 2)
.venv/bin/python -m scripts.tiny_circuit_exact --out validation_output/markov_tests/tiny

# Direct memory diagnostic on real g_eff(t) data (Markov test 4; needs a campaign from step 2
        above first)
.venv/bin/python -m scripts.markov_geff_test --dir validation_output/my_run --N 2000

# Delta^2(g_eff) vs. M convergence curve (reuses an existing campaign, no re-simulation)
.venv/bin/python -m scripts.geff_delta2_vs_m --dir validation_output/my_run --N 2000

# Dynamics-under-constant-drive figures (conductance traces, current, autocorrelation) with the
        working model
# (needs validation_output/fdtest/pilot_N2000_V2.5.npz from a prior `fdtest_pilot run 2000 2.5`
        call)
PYTHONPATH=. .venv/bin/python docs/report_figures/make_dynamics_figures.py
```

`docs/on_hold/validate_p_decay/validate_p_decay.py` uses an early, mis-scaled model (double-applies Δt to the switching probabilities) superseded on 2026-09-18 — do not run it; see its own `README.md` and `validation_output/old_model/README.md`. It was moved out of `scripts/` for exactly this reason.

Building the progress report PDFs

`md_to_pdf.sh` resolves the figures' image paths relative to the current directory, so it must be run **from inside `docs/`**, not the repo root — running it from the root silently produces a PDF with every figure missing (warnings only, exit status 0):

```
cd docs
bash md_to_pdf.sh REPORT_FDTest_progress.md
bash md_to_pdf.sh REPORT_FDTest_progress_brief.md
```

Claude Code workflow files

`memory/`, `RESUME_FROM_HERE.md`, and `docs/superpowers/` are continuity artifacts for Claude Code sessions working on this project across time — a persistent memory store, a "resume with zero context" handoff document, and design specs/plans from past work sessions,

respectively. None of them is needed to install, run, or understand the simulator or the FDTest; skip them unless you're picking up the project inside a Claude Code session.